Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/26/2005
PI Name:	Zimmerli, Gregory Ph.D.		
Project Title:	Confocal and Two-Photon Microscopy for the Assessment of Countermeasures in Bone Loss and Immunology		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Gregory.A.Zimmerli@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	216-433-6577
Organization Name:	NASA Glenn Research Center		
PI Address 1:	21000 Brookpark Road		
PI Address 2:	MS 110-3		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	11/01/2003	**End Date:**	07/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Zimmerli, Greg	**Contact Phone:**	216-433-6577
Contact Email:	Gregory.A.Zimmerli@grc.nasa.gov		
Flight Program:			
Flight Assignment:	Original end date was 12/31/2005.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fischer, David (NASA GRC) Knothe Tate, Melissa (Cleveland Clinic Foundation)		
Grant/Contract No.:	In-House		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We are developing fluorescence microscopy techniques such as two-photon Fluorescence Lifetime Imaging Microsocpy (FLIM) and Fluorescence Resonance Energy Transfer (FRET) to investigate the physiological response of osteoblast cells to mechanical stimuli such as fluid flow and mechnical strain. Our goal is to use fluorescence microscopy techniques to study the in-vitro response of osteoblast cells to mechanical perturbations. Using fluorescent nanoprobes such as CFP-YFP cameleons and e-GFP expression as a reporter of gene expression, we hope to uncover the mechanosensory mechanism by which bone cells respond to external loads and produce new bone. In microgravity, astronauts lose bone mineral density at a rate of 1% per month in weight bearing bones such as the hip and spine. By understanding the mechanism by which bone cells respond to mechanical load, one may be able to design more effective countermeasures to prevent bone loss.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The accelerated bone loss experienced by astronauts is like a severe osteoporosis. Although the two mechanisms of bone loss may be fundamentally different, it is possible that treatment of bone loss in microgravity may also be effective as a countermeasure for bone loss in persons with osteoporosis.
Task Progress:	A key element in our work is to validate our hypothesis that under appropriate mechanical loading conditions the osteoblast cultures will exhibit higher collagen production and matrix mineralization. We have succeeded in producing bone nodules in vitro using the MC3T3-E1 cell line, using mineralizing media refreshed every 3.5 days and keeping the cells in culture for up to 3 weeks. Alizarin Red stain was used to positively identify the bone nodules. Our next goal is to use GFP as a reporter molecule of, for example, osteocalcin gene expression in the mineralizing cultures. Many calcium-mediated signaling pathways have been implicated in the response of osteoblasts to shear stress. Previous studies have shown that there is an increase in the concentration of intracellular calcium upon exposure of cells to fluid flow. Because calcium plays such an important role in bone cells, one of our main goals is to develop the fluorescence techniques necessary to monitor calcium in the cells. A common calcium sensitive fluorophore is Fura-2, which is a ratiometric AM ester that detects intracellular calcium, and served as our first trial. The Fura-2 dye yields a good fluorescence signal after a 30 minute incubation at room temperature, but after only 4 hours in the 37 C incubator, much of the Fura-2 dye had diffused out of the cells. Thus, Fura-2 is not suitable for long-term imaging so we are currently investigating “cameleon” FRET probes developed by Dr. Roger Tsien’s lab in 1997. We have transfected cameleon probes into MC3T3-E1 cells using Effectene (Qiagen). Once inside the cell, the construct does not “leak out”, making it superior to the Fura 2 dye. The cells are examined 24-48 hours post transfection. In initial transfections, we transiently transfected the cells. The next step is to perform stable transfections by linearizing the construct, transfecting into the cell, and using antibiotic selection to select for transfected cells. The Becker & Hickl SPC-830 has been incorporated into the two-photon imaging system, interfaced to the Olympus scan head, and the photons are collected out the side “laser tweezer” port of the Leica microscope using a Hamamatsu PMT. The system has been initially tested using a fluorescent lake of fluorescein in methanol, and also using fluorescent microspheres dyed with two different stains. We have also acquired some preliminary FLIM images of the transfected MC3T3 cells.
Bibliography Type:	Description: (Last Updated: 01/26/2005)
Presentation	Zimmerli, G. "Biophotonics and Bone Biology" Great Lakes Photonics Symposium; Cleveland, OH Jun-2004
Presentation	Zimmerli, G. "Biophotonics and Bone Biology" NASA Conference: Strategic research to enable NASA’s exploration systems; Cleveland, OH Jul-2004